

PRASHANT SHAH

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EDUCATION

Ramapo College of New Jersey

B.S. Data Science & Computer Science (Double Major), Minor in Mathematics

- GPA: 3.84/4.00 | ICPC NY Honorable Mention, Presidential Scholarship

Mahwah, NJ

Expected: May 2026

EXPERIENCE

Center for Food Action

Data Engineer

Mahwah, NJ

09/2025 – Present

- Mapped the full data lifecycle from volunteer-entry collection forms through storage to grant-reporting pipelines, identifying that data quality failures originated at the collection layer due to absent input validation and ambiguous field definitions, not in downstream analysis.
- Built a Flask and PostgreSQL system with structured validation rules and standardized field definitions, implementing bidirectional sync with PantrySoft's denormalized legacy database to eliminate manual reconciliation steps and reduce downstream reporting discrepancies.

Trimble Inc.

Software Engineering Intern

Princeton, NJ

05/2025 – 08/2025

- Developed production components in C# .NET within a multi-service enterprise platform processing data across distributed teams and system boundaries, writing SQL queries with multi-table joins to model configuration relationships and surface conflict patterns across production and development environments.
- Developed REST APIs integrating with TMW Suite enterprise transportation system to manage service configurations across environments, and built an authentication proxy using Azure Workload Identity Federation for role-based access control across multi-environment deployments.

Novartis Pharmaceuticals Corporation

Software Engineering Intern

East Hanover, NJ

06/2024 – 08/2024

- Designed PostgreSQL data models to centralize project information from multiple departmental sources in a regulated pharmaceutical environment where data integrity and compliance validation are mandated, writing SQL queries to support cross-functional reporting on resource allocation and proposal pipelines.
- Built Power BI and Tableau dashboards with Azure Functions data processing workflows enabling administrators to monitor budget summaries, team allocation, and approval status across 100+ projects, reducing manual reporting time from 2 days to hours.

RESEARCH & PROJECTS

Resolving Intent-Content Conflict in Retrieval | LLM, GNN, RAG | [Link](#)

05/2025 – Present

- Designed a heterogeneous knowledge graph encoding Ekman's six emotions as first-class traversable entities over MovieLens 25M, with Graph Transformer encoding and a hybrid scoring function weighting semantic similarity against emotional alignment, evaluated using RAGAS-based faithfulness metrics and an Emotion Leave-One-Out protocol.
- Identified that performance matched baselines not due to architectural limitations but untrained GNN encoder weights and paradigm mismatch between the affective retrieval task and the comparison baseline, traced through systematic ablation of the emotion signal path with causal confirmation via edge-removal interventions.

Personal Canvas Agent | TypeScript, Next.js, Vercel AI SDK, PostgreSQL, pgvector | [Link](#)

02/2026

- Built a multi-step tool-calling AI agent with 9 typed tools backed by parameterized SQL against a Canvas LMS-mirrored PostgreSQL schema, persistent memory via pgvector embeddings, and a reactive planner that diffs state on every request to inject live context into the system prompt. Won HackRamapo.
- Designed session persistence with per-session write locks to prevent race conditions and automatic context compaction via a secondary LLM call, keeping the agent's context window focused across long conversations without losing history.

SKILLS

Languages: Python, C#, TypeScript, JavaScript, Java, C/C++, SQL (PostgreSQL, MySQL), Bash

Frameworks & Libraries: React, Node.js, .NET, Flask, Express, PyTorch, Pandas, scikit-learn

Cloud & Infrastructure: Azure, AWS, Docker, Git, GitHub Actions, CI/CD, Linux

Practices: Agentic Development (Claude Code, Cursor, Codex), REST APIs, Agile, Code Review, Testing